Git Push

TOPIC Greedy Algo for k-means DATE

	x_1	x_2
S_1	1	1
S_2	1	2
S_3	2	2
S_4	8	8
S_5	8	9
S_6	9	8

lets say $k = 2$

So greedy approach.

k-means objective function

$$\min_{C_1, C_2 \dots C_k} \left\{ \sum_{k=1}^{K} \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^{d} \left(x_j^{(i)} - x_j^{(i')} \right)^2 \right\}$$

but this is computationally complex

1) Randomly put samples in k-clusters

C_1: $S_1(1,1)$, $S_3(2,2)$, $S_4(8,8)$

C_2: $S_2(1,2)$, $S_5(8,9)$, $S_6(9,8)$

2) Calculate the centroids

Centroid $Q_1 = \left(\frac{1+2+8}{3}, \frac{1+2+8}{3} \right)$

$Q_1 = \left(\frac{11}{3}, \frac{11}{3} \right)$

Centroid $Q_2 = \left(\frac{1+8+9}{3}, \frac{2+9+8}{3} \right)$

$Q_2 = \left(\frac{18}{3}, \frac{19}{3} \right)$

3) distance to centroid

$(1,1)$ S1 $Q_1 \left(\frac{11}{3}, \frac{11}{3}\right)$

$R_1^1 = \sqrt{\left(1 - \frac{11}{3}\right)^2 + \left(1 - \frac{11}{3}\right)^2}$

$= \sqrt{\left(\frac{8}{3}\right)^2 + \left(\frac{8}{3}\right)^2}$

$= \sqrt{2\left(\frac{8}{3}\right)^2} = 3.771$ ✓

S1 to C1

$Q_2 \left(\frac{18}{3}, \frac{19}{3}\right)$

$R_1^2 = \sqrt{\left(1 - \frac{18}{3}\right)^2 + \left(1 - \frac{19}{3}\right)^2}$

$= \sqrt{\left(\frac{15}{3}\right)^2 + \left(\frac{16}{3}\right)^2}$

$= \sqrt{53} = 7.31$

$(1,2)S2$ $Q1\ (\frac{11}{3}, \frac{11}{3})$

$R_2^1 = \sqrt{(1-\frac{11}{3})^2 + (2-\frac{11}{3})^2}$

$= \sqrt{(\frac{8}{3})^2 + (\frac{5}{3})^2}$

$= \sqrt{7.11 + 2.77} = 3.14$

S2 belongs in C_1

$Q2\ (\frac{18}{3}, \frac{19}{3})$

$R_2^2 = \sqrt{(1-\frac{18}{3})^2 + (2-\frac{19}{3})^2}$

$\sqrt{(\frac{15}{3})^2 + (\frac{13}{3})^2}$

$\sqrt{25 + 18.77} = 6.6144$

$(2,2)S3$ $Q1\ (\frac{11}{3}, \frac{11}{3})$

$R_3^1 = \sqrt{(2-\frac{11}{3})^2 + (2-\frac{11}{3})^2}$

$= \sqrt{2.77 + 2.77}$

$= 3.32$

S3 belong in C_1

$Q2\ (\frac{18}{3}, \frac{19}{3})$

$R_3^2 = \sqrt{(2-\frac{18}{3})^2 + (2-\frac{19}{3})^2}$

$= \sqrt{(\frac{12}{3})^2 + (\frac{13}{3})^2}$

$= \sqrt{16 + 18.77} = 5.89$

$(8,8)S_4$ $Q_1\ (\frac{11}{3}, \frac{11}{3})$

$R_4^1 = \sqrt{(8-\frac{11}{3})^2 + (8-\frac{11}{3})^2}$

$= \sqrt{(8-3.7)^2 + (8-3.7)^2}$

$= 6.08$

S4 belong in C_2

$Q_2\ (\frac{18}{3}, \frac{19}{3})$

$R_4^2 = \sqrt{(8-\frac{18}{3})^2 + (8-\frac{19}{3})^2}$

$= \sqrt{(8-6)^2 + (8-6.33)^2}$

$= \sqrt{4 + 2.78}$

$= 2.6$

$(8,9)S_5$ $Q_1\ (\frac{11}{3}, \frac{11}{3})$

$R_5^1 = \sqrt{(8-\frac{11}{3})^2 + (9-\frac{11}{3})^2}$

$= \sqrt{(8-3.7)^2 + (9-3.7)^2}$

$= \sqrt{18.49 + 28.09}$

$= 6.82$

S5 belong in C_2

$Q_2\ (\frac{18}{3}, \frac{19}{3})$

$R_5^2 = \sqrt{(8-\frac{18}{3})^2 + (9-\frac{19}{3})^2}$

$= \sqrt{(8-6)^2 + (9-6.33)^2}$

$= \sqrt{4 + 7.12}$

$= 3.33$

$(9,8)S_6$ $Q_1\ (\frac{11}{3}, \frac{11}{3})$

$R_6^1 = \sqrt{(9-\frac{11}{3})^2 + (8-\frac{11}{3})^2}$

$= \sqrt{(9-3.7)^2 + (8-3.7)^2}$

$= \sqrt{28.09 + 18.49}$

$= 6.82$

S6 belong in C_2

$Q_2\ (\frac{18}{3}, \frac{19}{3})$

$R_6^2 = \sqrt{(9-\frac{18}{3})^2 + (8-\frac{19}{3})^2}$

$= \sqrt{(9-6)^2 + (8-6.33)^2}$

$= \sqrt{9 + 2.78}$

$= 3.43$